Unified Modeling Language (UML)

Presented by

Roaa Ayman

Contents



Introduction to UML 01

Use Case Diagrams 02

Example Use Case: Bank ATM System 03

Example Use Case: Library Management System 04

Additional Use Case Examples 05

Task 06



01 Introduction to UML

What is UML?

01

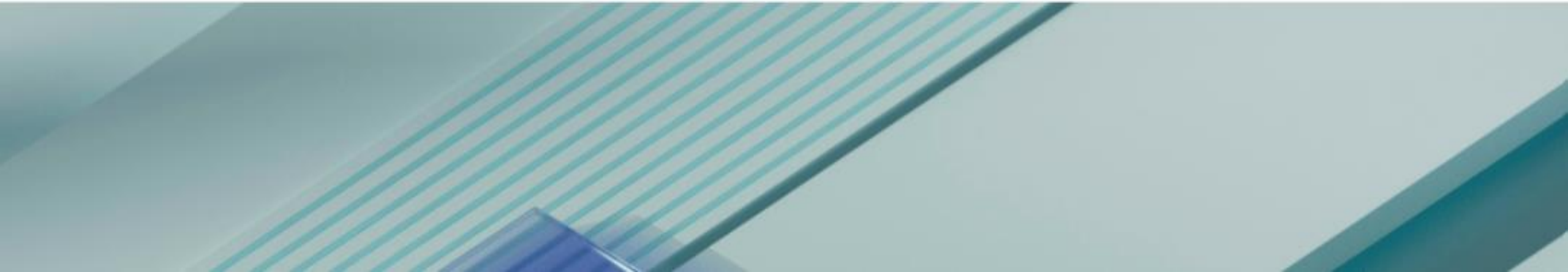
Definition

UML is a graphical language for visualizing, specifying, constructing, and documenting software-intensive system artifacts.

02

Goals of UML

Provide users with a ready-to-use, expressive visual modeling language so they can develop and exchange meaningful models.



UML Diagrams Overview

Types of UML Diagrams

- Use Case Diagram
- Activity Diagram
- Class Diagram
- Communication Diagram
- Component Diagram
- Deployment Diagram
- Object Diagram
- Package Diagram
- Sequence Diagram
- State Diagram



UML Diagrams

UML Diagram	Definition
Use Case Diagram	Shows the interaction between users (actors) and the system, focusing on system functionality.
Activity Diagram	Represents the workflow or activities in a process, showing the sequence of steps.
Class Diagram	Shows the structure of the system by displaying classes, attributes, methods, and relationships.
Communication Diagram	Focuses on the interactions between objects and how they collaborate through messages.
Component Diagram	Shows how the system is divided into components (modules) and the dependencies between them.
Deployment Diagram	Displays the physical deployment of software components on hardware nodes.
Object Diagram	Represents objects at a particular moment in time, showing their states and relationships.
Package Diagram	Organizes related classes and elements into packages to simplify system structure.
Sequence Diagram	Shows how objects interact in a time sequence, emphasizing the order of messages.
State Diagram	Describes the states of an object and how it transitions from one state to another.



02
Use Case Diagrams

Basics of Use Case Diagrams

01

Definition of Use Case

A set of scenarios describing interaction between a user and a system.

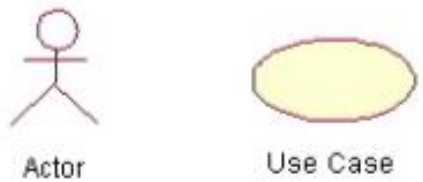
Displays relationships among actors and use cases.

02

Main Components

Actors: persons, organizations, or external systems interacting with the system.

Use Cases: sequences of actions delivering value to actors.



Identifying Actors and Use Cases



Actor Identification Questions

Who uses the main system functionalities (primary actors)?
Who maintains or administers the system (secondary actors)?
Required hardware and external system interactions?
Interested parties in system results?



Use Case Identification Questions

What functions does the actor need?
Does the actor read, create, modify, or store data?
Does the actor receive system event notifications?

Actor Identification Questions

Question	Explanation	Example - (ATM System)
Who uses the main system functionalities? (Primary Actors)	The people or systems that directly interact with the main features of the system to achieve their goals.	**Customer** – uses the ATM to withdraw, deposit, or check balance.
Who maintains or administers the system? (Secondary Actors)	The people or systems responsible for setup, maintenance, or administration tasks.	**ATM Operator / Bank Admin** – maintains the ATM, loads cash, updates records.
Required hardware and external system interactions	The physical devices and external systems that the main system depends on to work properly.	**Hardware:** Card reader, cash dispenser, printer, network connection. **External Systems:** Bank server, customer database.
Interested parties in system results	Stakeholders who care about the system's output or performance, even if they don't use it directly.	**Bank management, auditing department, customer service** – monitor reports, transactions, and performance.

Use Case Identification Questions

Question	Explanation	Example - (ATM System)
What functions does the actor need?	Describes what tasks or operations the actor wants to perform in the system.	**Customer**: Withdraw cash, check balance, transfer funds.ATM Operator: Refill cash, perform maintenance.
Does the actor read, create, modify, or store data?	Defines how the actor interacts with system data — reading, entering, editing, or saving it.	**Customer:** Reads balance, creates transaction records. **Admin:** Modifies user accounts, updates cash inventory.
Does the actor receive system event notifications?	Specifies whether the system sends alerts or messages to the actor about certain events or actions.	**Customer:** Receives error messages, transaction receipts. **Admin:** Receives alerts about machine errors or low cash levels.



03
Example Use Case: Bank ATM System

ATM Components & Functionalities

- A bank has several Automated Teller Machines (ATMs), which are geographically distributed and connected via a wide area network to a central server.

Customer Transactions

1- Insert Card / Authenticate
2- Withdraw cash
3- Query account balance.
4- Transfer funds between accounts.
5- Cancel Transaction

ATM Hardware

1- Card Reader
2- Cash Dispenser
3- Keyboard/Display
4- Receipt Printer.

Transaction Flow

Step	Description
1. Start Transaction	Customer inserts ATM card. The system reads card number, start date, and expiration date.
2. Card Validation	System checks: - Expiration date not passed. - PIN matches stored PIN. - Card not lost or stolen. - Maximum 3 attempts allowed, then card is confiscated.
3. Choose Transaction	Customer selects one option: - Withdraw cash. - Query balance. - Transfer funds.
4. Withdrawal Check	System verifies: - Sufficient account funds. - Daily limit not exceeded. - Enough cash in dispenser.
5. Transfer Check	System verifies: - Customer has at least two accounts. - Sufficient funds in debit account.
6. Transaction Approval	If approved: - Withdraw → dispense cash, print receipt, eject card. - Query/Transfer → print receipt, eject card.
7. Cancel Transaction	Customer can cancel anytime. The transaction ends and the card is ejected.
8. Records Maintenance	Bank server stores: - Customer records. - Account records. - Card records.
9. ATM Operator Role	Operator can: - Start up ATM. - Shut down ATM. - Refill cash. - Perform maintenance.

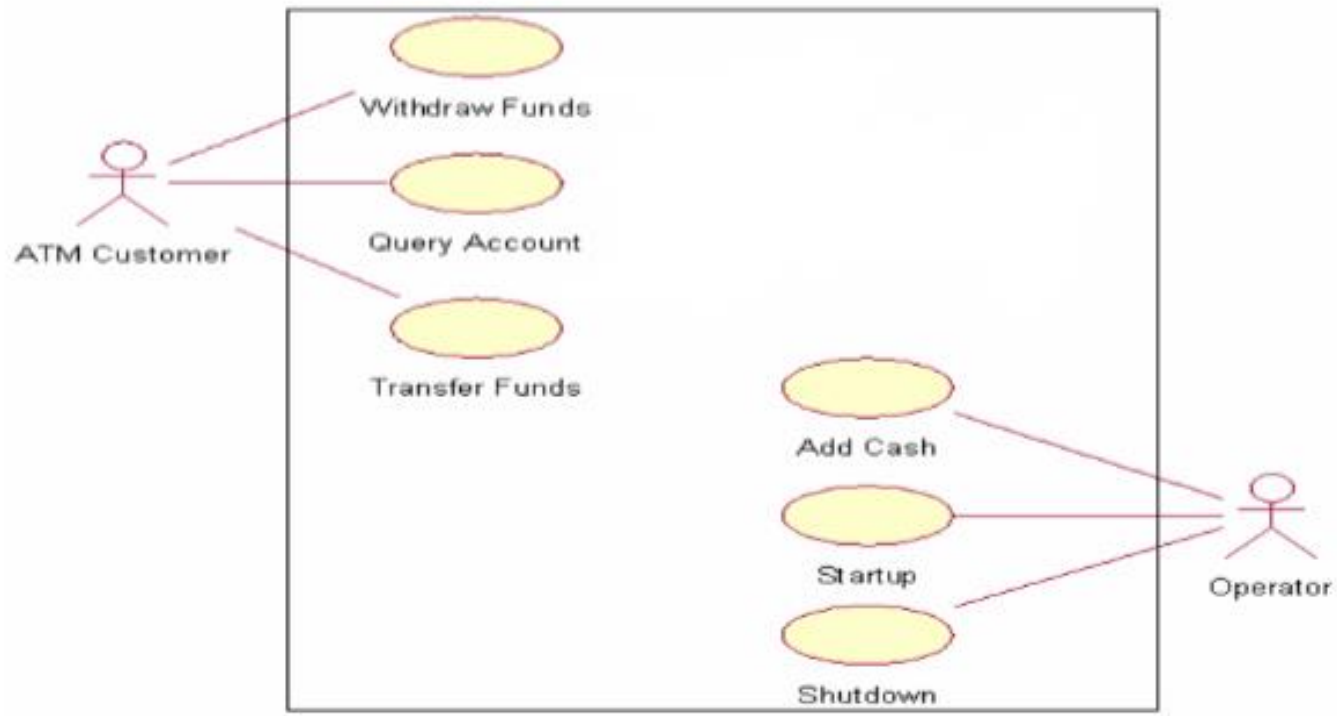
ATM - Use Case Model Summary

01

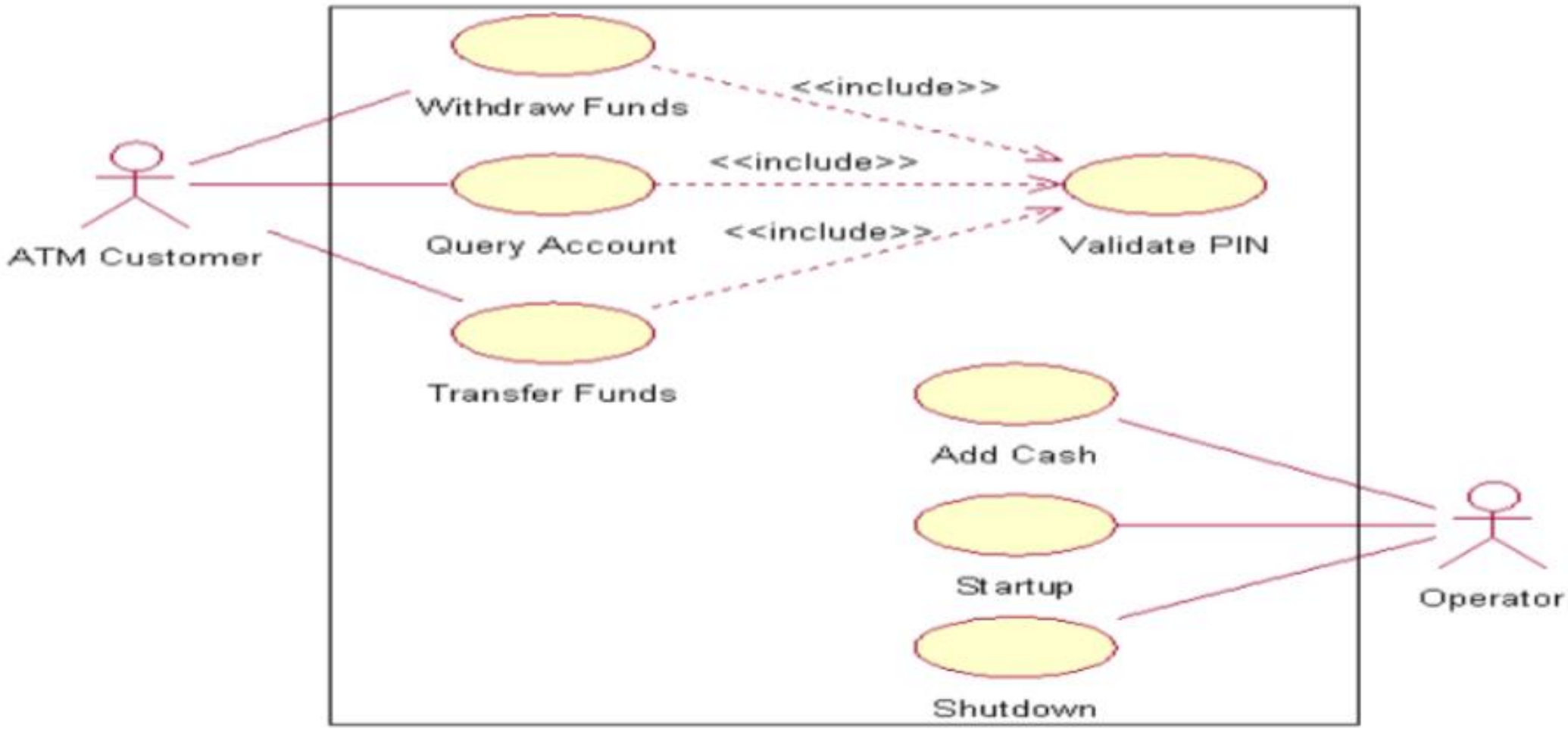
Withdraw Funds, Query Account, Transfer Funds (customer roles).

02

Add Cash, Startup, Shutdown (operator roles).



ATM: Use Case Model



Include and Extend

Concept	meaning	*example*	notes
Include	The main use case must always perform the included use case. It is mandatory.	*Borrow Book* → includes → *Check Availability*	The included use case cannot stand alone; it's always part of the main use case.
Extend	The extension use case happens only sometimes (optional or conditional).	*Return Book* → extends → *Calculate Fine*	The extended use case is triggered only if certain conditions are met (e.g., book returned late).



04
Example Use Case: Library Management System

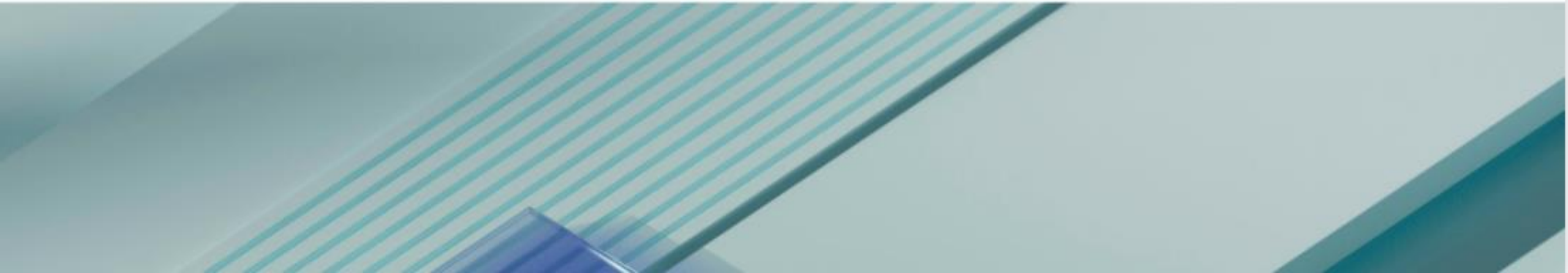
Library System Overview

01

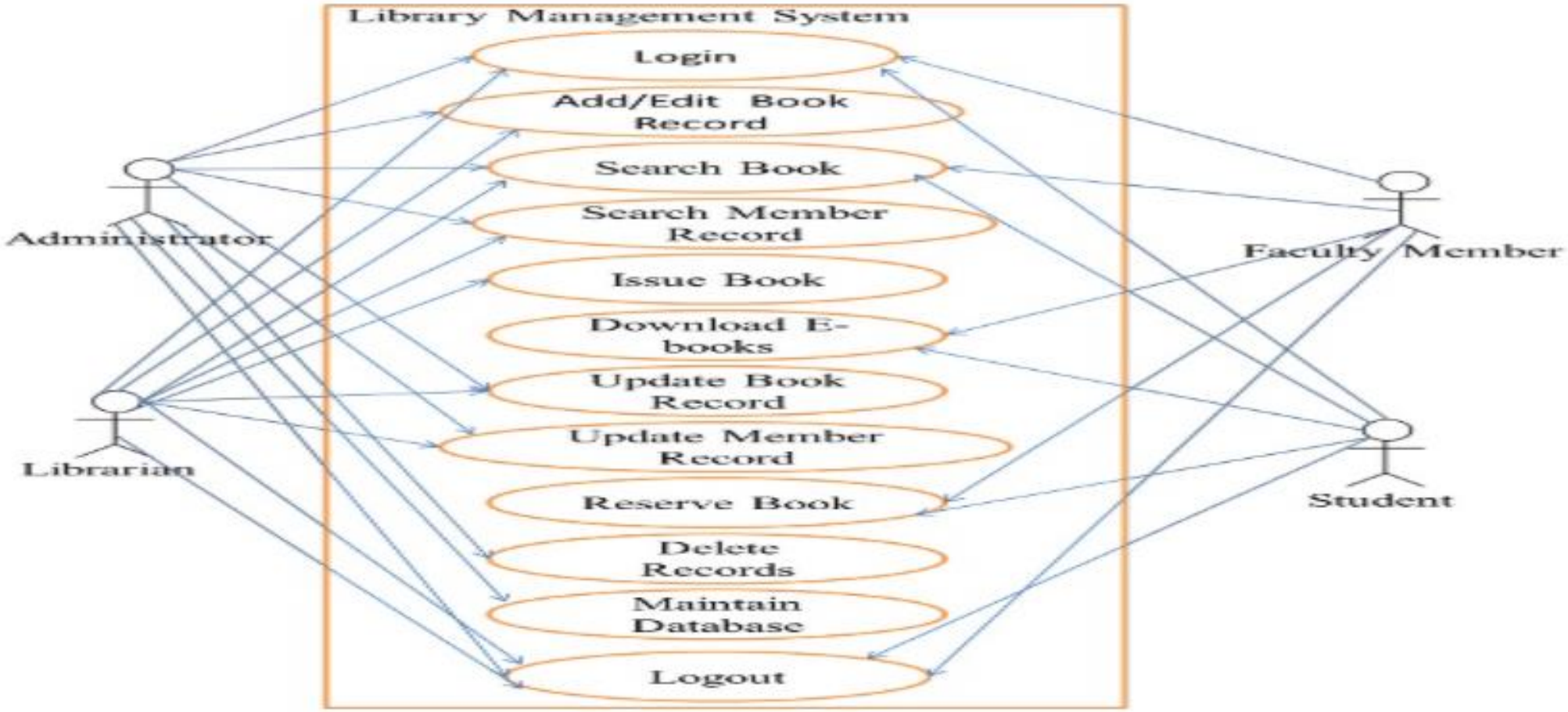
Stores book details and supports book add/search, borrow, return functionality.

02

Users: Administrator, Staff, Students.



Library Management system use case



Use Case: Login

Step1

Actors

Admin, Student, Librarian, Staff.

Step2

Flow

Insert username and password; login successful with correct credentials.
Alternative flows handle incorrect username or password.

Use Case ID	1
Use Case Name	Login
Actors	Admin, student, librarian, staff
Preconditions	• Admin, student, librarian must do registration
Post conditions	login successfully
Normal Flow	1- Insert user name
	2- Insert password
	3- Login
Alternative Flows	in step 1 : insert the incorrect user name in step 2 : insert the incorrect password

Use Case: Search

Use case ID	3
Use case name	Search
Primary Actors	Administrator, Librarian, Faculty Member, Students
Pre-condition	The book or student to be searched should have been registered in the database of the library management system.
Main Scenario	1. The student, staff, librarian, or administrator logs into the system. 2. If the login is successful, the student, staff, librarian, or administrator enters the book name, ISBN, or author name and presses search. 3. If the search is successful, then that book is displayed on the screen. 4. To search for a student, the administrator or staff logs into the system. 5. If the login is successful, it is possible to search for any student by entering their ID. 6. To search for a staff member, the administrator enters their login ID. 7. If successful, they can search for any of the staff members.
Alternate Scenarios	1. The login fails. 2. The student, librarian, staff, or administrator can re-register themselves. 3. If the search is unsuccessful, the administrator should add that member. 4. If the book search is unsuccessful, then that book should be added.



05
Additional Use Case Examples

Additional Use Case Examples

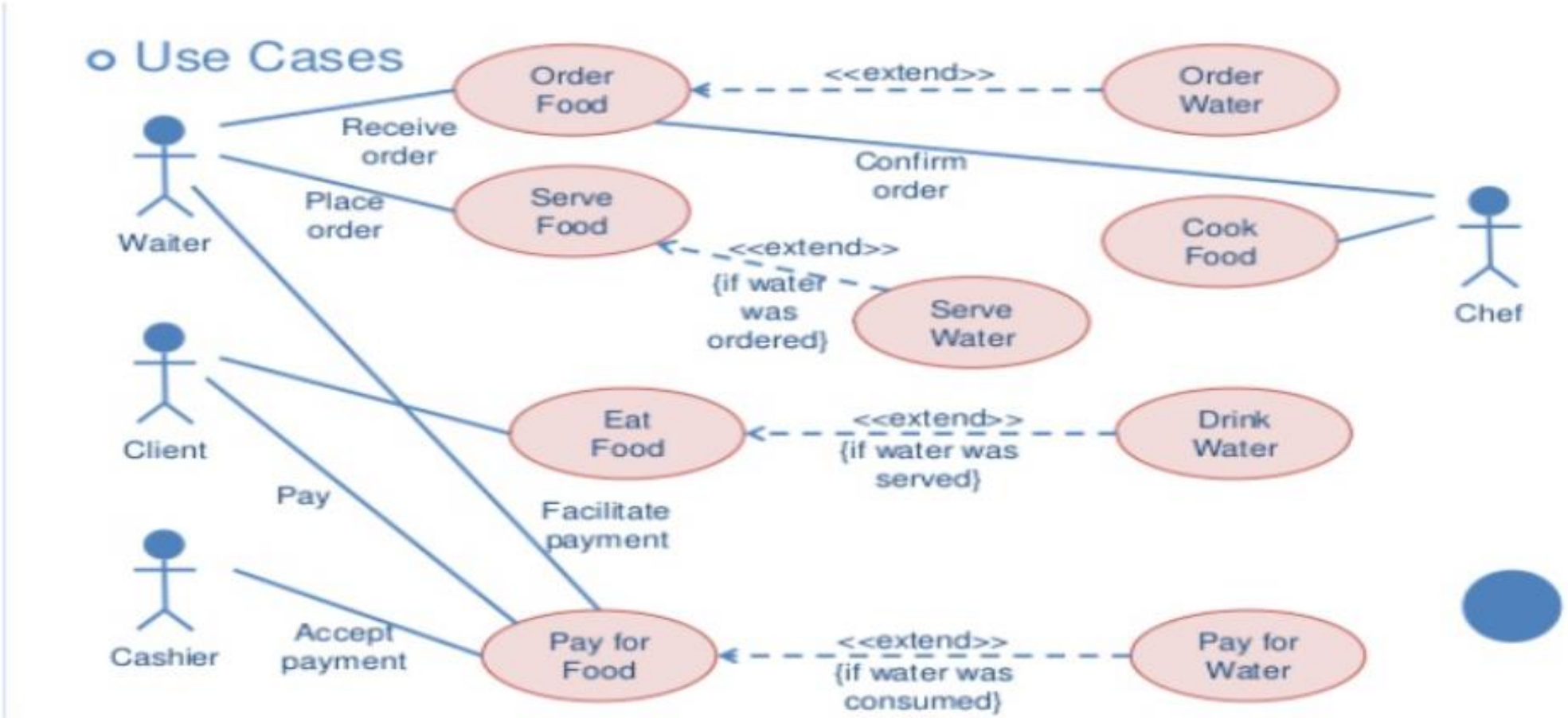
Restaurant Use Case

Order, Order Food/Water, Serve order, Cook, Waiter activities.
Extend relationships for payment and drink consumption.

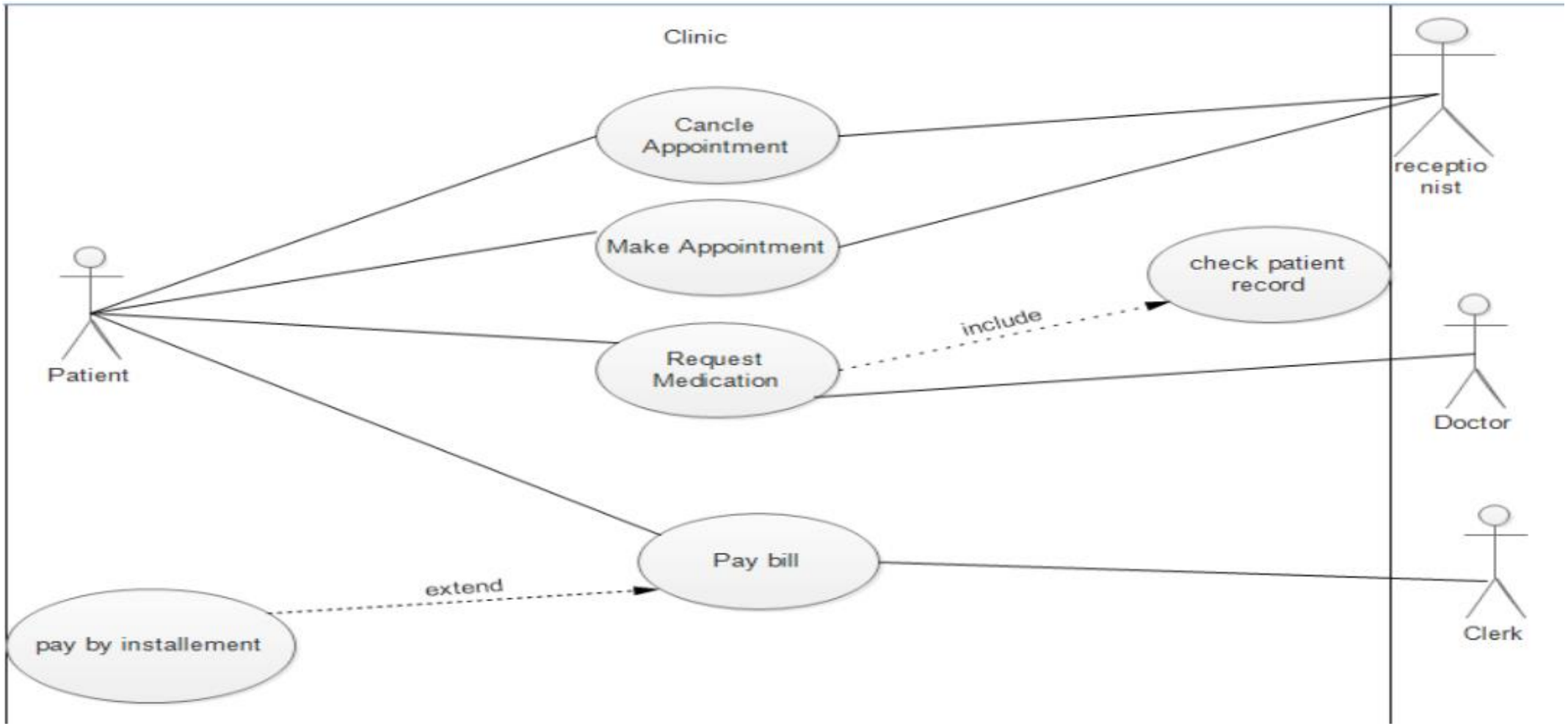
Clinic Use Case

Make/Cancel Appointment, Check Patient Record, Request Medication, Pay Bill.
Payment extension via clerk with installment option.

Restaurant Use case



Clinic Use case




06
Task
Technology

Task : Student Registration System

“

Scenario:

- Students register for courses each semester using a university system.
- Draw a **Use Case Diagram** that shows how students and admin interact with the system.
- Create a **Use Case Table** for this scenario.
- **Hint:** Actors: *Student*, *Admin*
- Use cases: *Login*, *Register Course*, *View Grades*, *Manage Students*.

Thanks

Presented by
Roaa Ayman